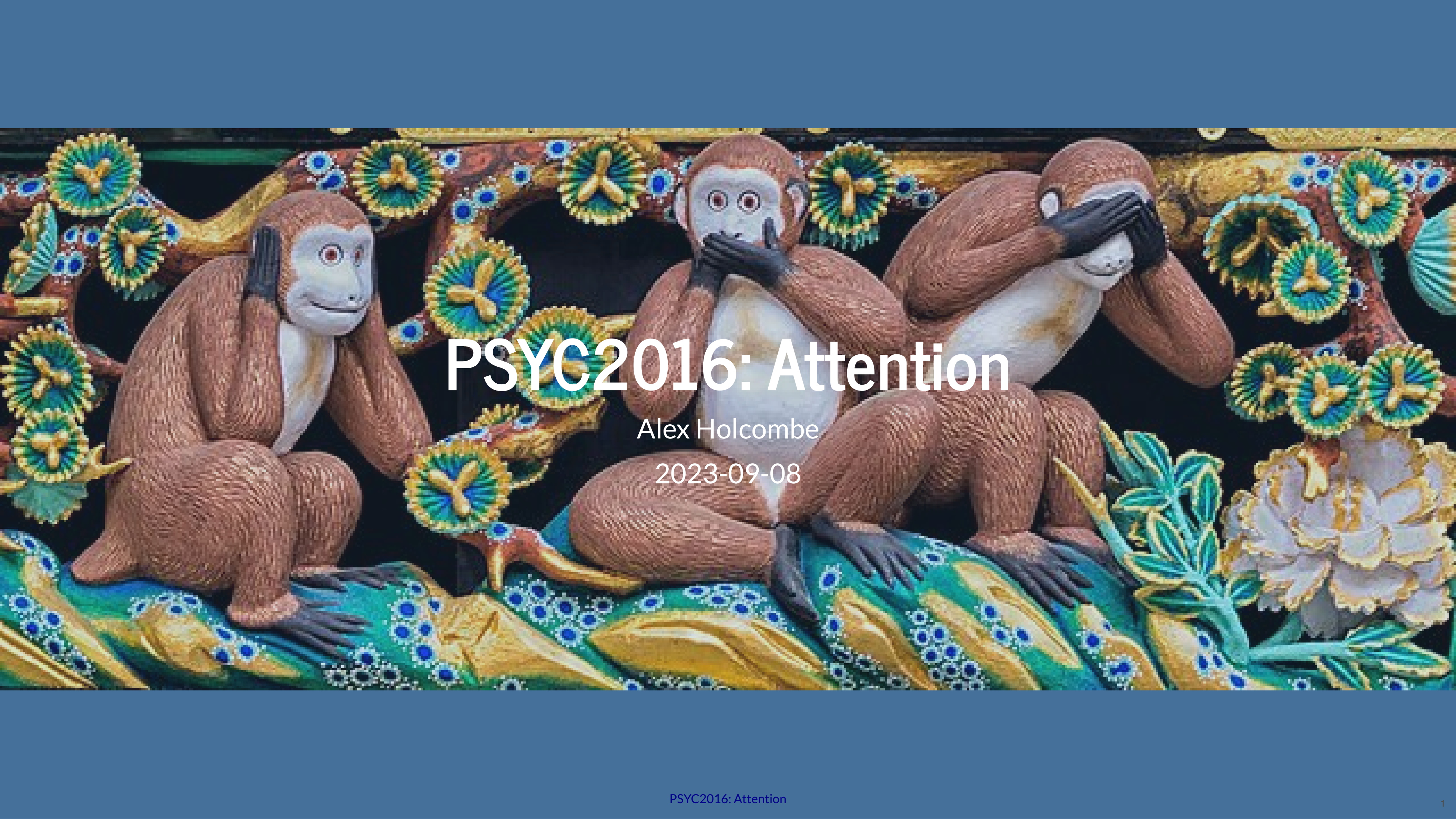




PSYC2016: Attention

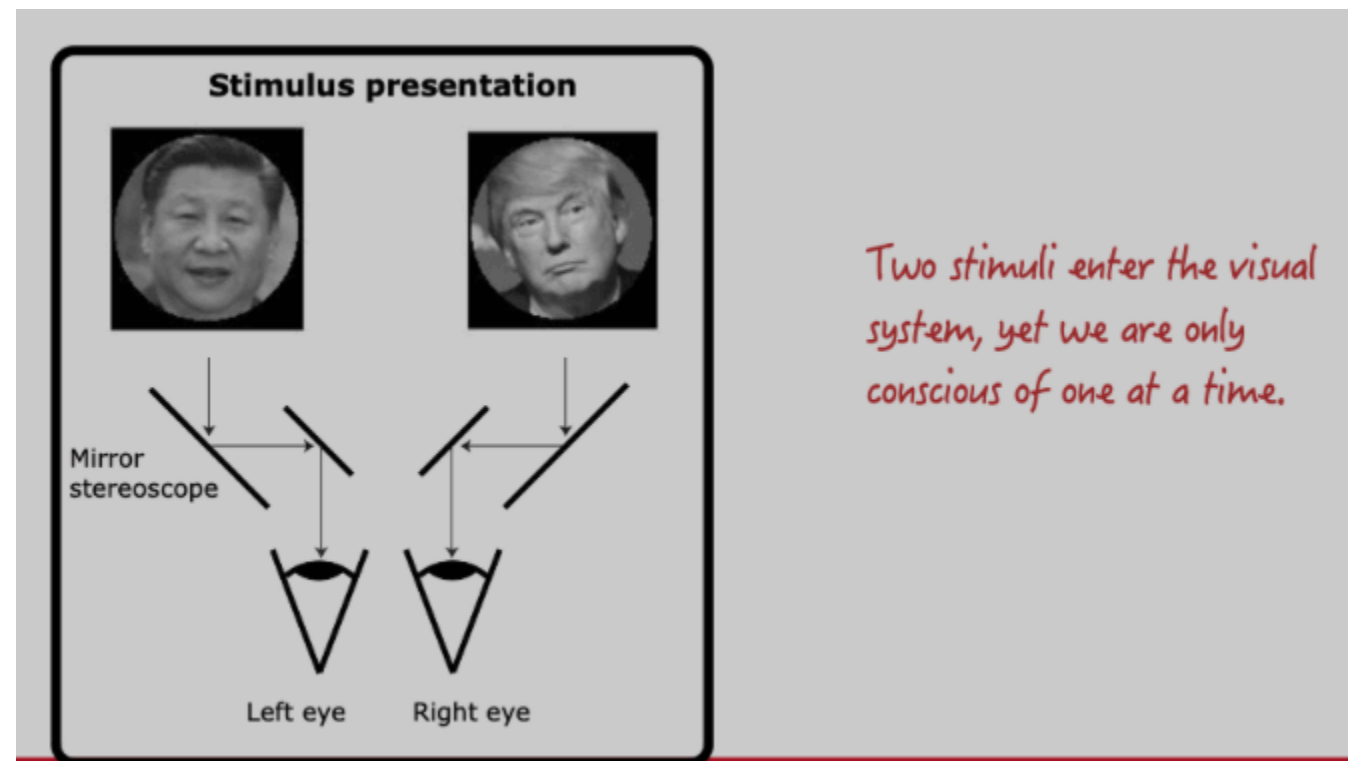
Alex Holcombe

2023-09-08

You said the two letters are processed simultaneously, but other lecturer said the faces of Trump and Jinping alternate



You said the two letters are processed simultaneously, but other lecturer said the faces of Trump and Jinping alternate



Binocular rivalry versus dual letters

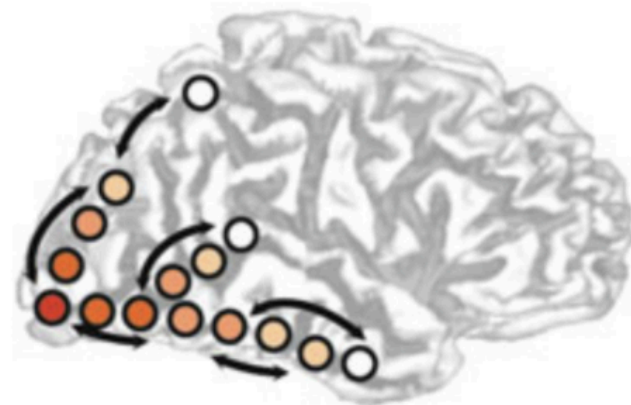
- Binocular rivalry: different stimuli presented to the two eyes
 - Processed in parallel to a point
 - *Incompatible*: brain expects the same object in both eyes, therefore
 - Direct competition for a single location, and winner occurs relatively early
- Dual letters in different locations
 - No incompatibility, so *processed in parallel until bottleneck*, then compete for that

Didn't understand "Brain loop cycle bits"

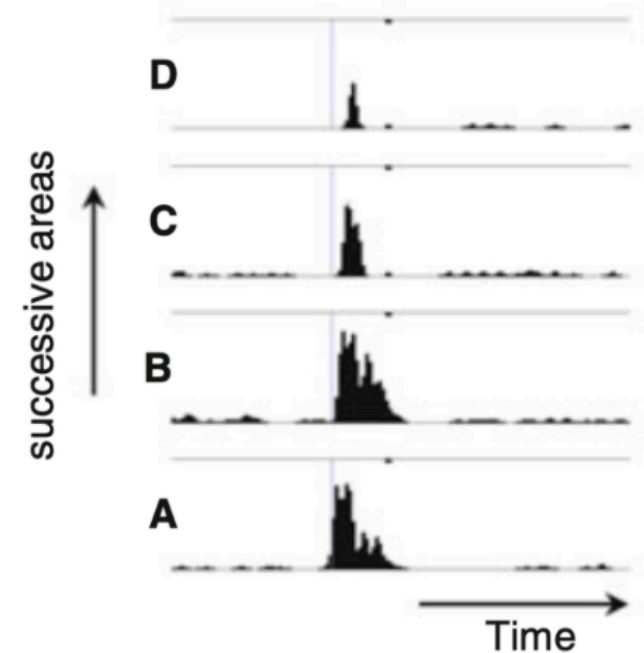
Global neuronal workspace theory

Feedforward visual processing

The processing of visual information in the feedforward direction is thought to remain subliminal or, at least, inaccessible to further cognitive processing.

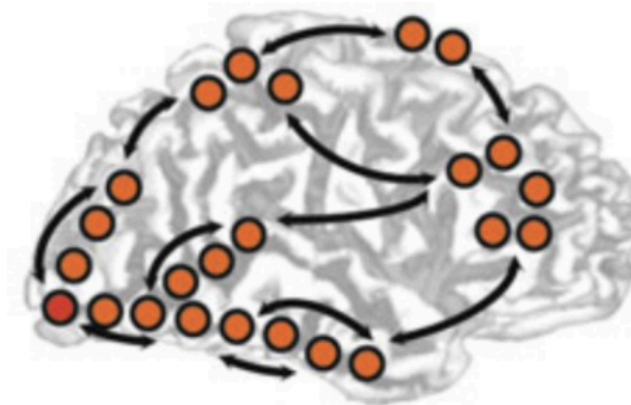


b Propagation with failure of ignition

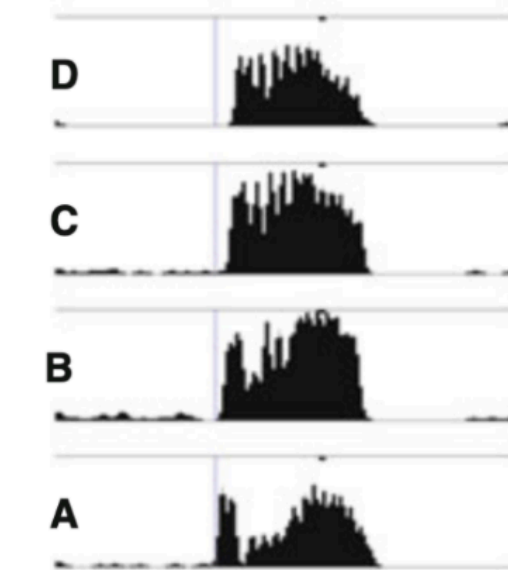


Activated global neuronal workspace

Once a signal triggers the frontal cortex, a network reverberation is thought to allow visual representation to be both conscious and available to other cognitive systems.



Global workspace ignition



(Dehaene and Changeux 2005).

- There's a bottleneck prior to the frontal lobe
- If a stimulus gets through, it results in reverberating widespread activity associated with that stimulus

Consciousness



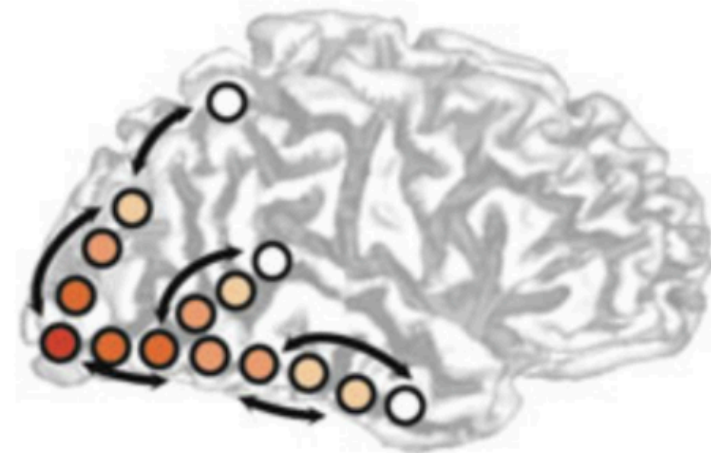
What happens when a stimulus location is selected?

- Activity makes it through the bottleneck
- Recurrent activity

What happens when a spatial location is selected?

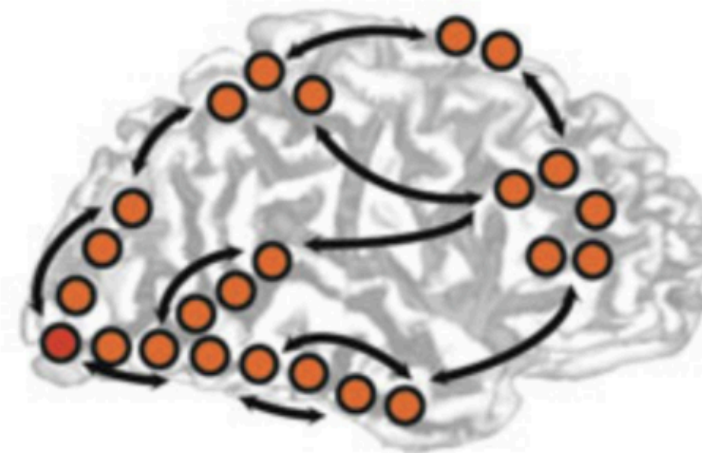
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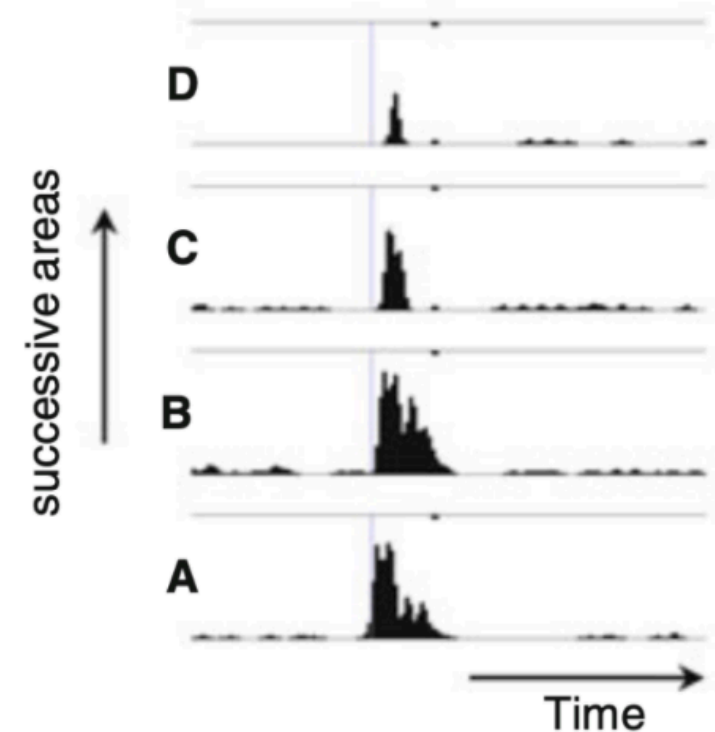


Activated global neuronal workspace

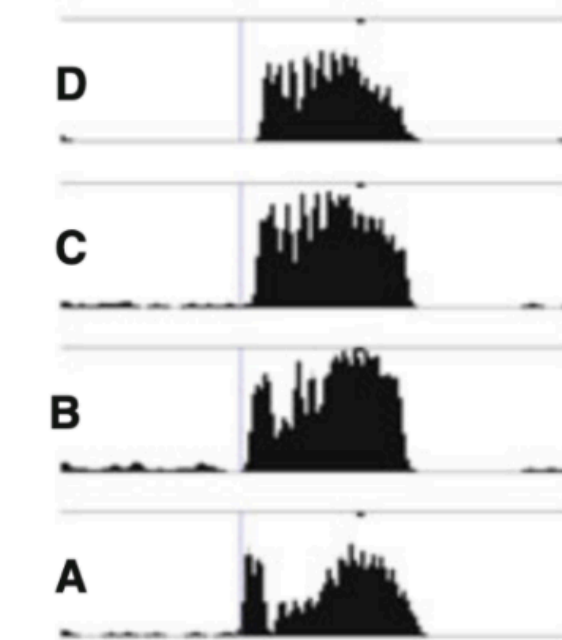
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Global workspace ignition

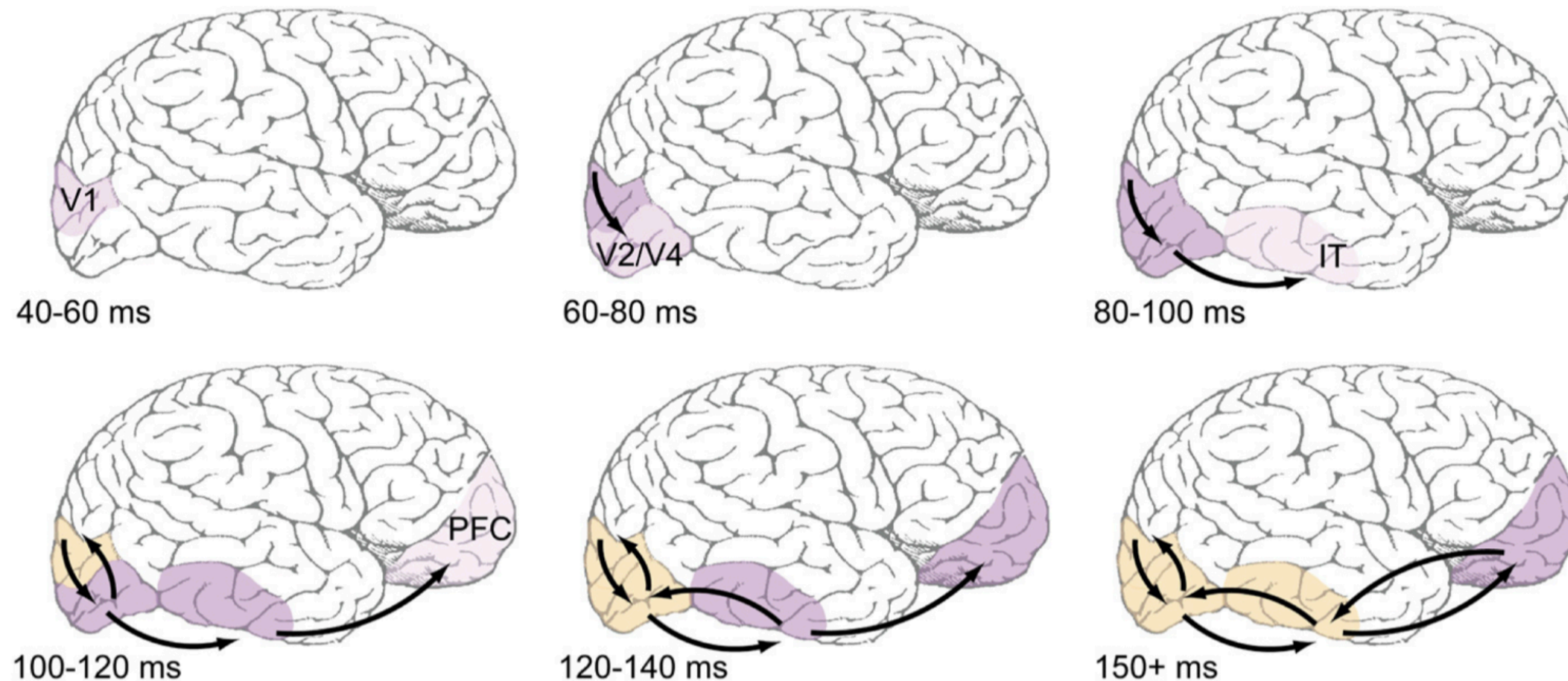


(Dehaene and Changeux 2005).

Why can't we be conscious of all stimuli?

What happens when a spatial location is selected?

Consciousness is not a place but a process of widespread recurrent activity



::: notes

Recurrent activity :::